CS229 Machine Learning

Computational Problems
Preparation for the Final Exam

Instructions:

- This is a preparation for the **open-book** final exam.
- Preparation consists of short computational problems covering Chapters 1–7 of the PRML textbook.
- Each problem should be solved in max 30 minutes with minimal aids like a basic calculator.
- Provide your answers in a clear, step-by-step manner to demonstrate your reasoning.

Spring 2026

Computing Basic Statistics (Mean, Covariance)

Statement. We have three data points in 2-dimensional space:

$$\mathbf{x}_1 = (1, 2), \quad \mathbf{x}_2 = (2, 3), \quad \mathbf{x}_3 = (3, 5).$$

(a) Compute the sample mean vector of these points.

(b) Compute the 2×2 sample covariance matrix.

(c) Verify that the covariance matrix is positive semidefinite by performing an eigenvalue analysis.

Grading:

- *Sample mean – 3 pts*
- *Covariance matrix – 4 pts*
- *PSD check (eigenvalue analysis) – 3 pts*

Total: 10 points

Maximum Likelihood for a Univariate Gaussian

Statement. We have a one-dimensional dataset:

$$\{2.0,\ 2.5,\ 3.5,\ 4.0\}.$$

We assume it comes from a Gaussian $\mathcal{N}(\mu, \sigma^2)$. We want to:

(a) Write down the log-likelihood function.

(b) Compute the maximum-likelihood estimates (MLE) $\hat{\mu}$ and $\hat{\sigma}^2$.

Grading:

- *Log-likelihood expression – 5 pts*
- *MLE for μ, σ^2 – 5 pts*

Total: 10 points

Evaluating a Simple Bayesian Update (Bernoulli-Beta)

Statement. We have a Bernoulli variable with parameter θ. Its prior is $\mathrm{Beta}(1, 1)$. We observe 2 successes and 1 failure. We want to:

(a) Write the posterior in Beta form.

(b) Give the new parameters α, β.

(c) Compute the posterior mean.

Grading:

- *Posterior formula – 3 pts*
- *Posterior parameters – 3 pts*
- *Posterior mean – 4 pts*

Total: 10 points

Constructing a Polynomial Design Matrix

Statement. We have three 1D input–output pairs:

$$(x_1, t_1) = (0, 1), \quad (x_2, t_2) = (1, 2), \quad (x_3, t_3) = (2, 5).$$

We fit a polynomial of degree 2:

$$y(x) = w_0 + w_1 x + w_2 x^2.$$

(a) Construct the design matrix $\mathbf{\Phi}$.

(b) Write the normal equations for $\mathbf{w}$.

(c) Solve for w_0, w_1, w_2.

Grading:

- *Design matrix – 5 pts*
- *Normal equations – 3 pts*
- *Solve for parameters – 7 pts*

Total: 15 points

Regularized Polynomial Regression with a Small Dataset

Statement. We have 5 data points in 1D:

$$(x_1, t_1) = (0,\ 0.2),\ (x_2, t_2) = (1,\ 1.1),\ (x_3, t_3) = (2,\ 1.4),\ (x_4, t_4) = (3,\ 2.6),\ (x_5, t_5) = (4,\ 4.5).$$

We fit a polynomial of degree 2,

$$y(x) = w_0 + w_1 x + w_2 x^2,$$

with regularization parameter λ. That is, we minimize

$$\sum_{n=1}^{5} \big[t_n - y(x_n)\big]^2 \ + \ \lambda \sum_{j=0}^{2} w_j^2.$$

(a) Construct the 5×3 design matrix $\mathbf{\Phi}$.

(b) Explain how to form the ridge solution.

(c) Suppose $\lambda = 0.5$. Estimate $\mathbf{w}$ and discuss whether it over-regularizes or not.

Grading:

- *Design matrix – 5 pts*
- *Ridge matrix formation – 5 pts*
- *Approx* $\mathbf{w}$ *and discussion – 5 pts*

Total: 15 points

Bias-Variance Decomposition at a Single Test Point

Statement. We are evaluating a regression model at a specific test input $\mathbf{x}_*$. The true underlying function yields $h(\mathbf{x}_*) = 2.0$, and the intrinsic target noise variance is $\epsilon^2 = 0.1$.

We train our model on three different random datasets $\mathcal{D}_1, \mathcal{D}_2, \mathcal{D}_3$. The corresponding predictions made by these trained models at the test point are:

$$y^{(1)}(\mathbf{x}_*) = 1.5, \quad y^{(2)}(\mathbf{x}_*) = 2.0, \quad y^{(3)}(\mathbf{x}_*) = 2.5$$

(a) Calculate the expected prediction $\bar{y}(\mathbf{x}_*)$ across the datasets.

(b) Calculate the squared bias at this test point.

(c) Calculate the variance of the predictions at this test point.

(d) Calculate the total expected squared loss $\mathbb{E}[L]$ at $\mathbf{x}_*$.

Grading:

- *Expected prediction – 2 pts*
- *Squared bias – 3 pts*
- *Variance – 3 pts*
- *Total expected loss – 2 pts*

Total: 10 points

Bayesian Linear Regression Predictive Distribution

Statement. Suppose we have 1D data $\{(0,1),(1,2),(2,5)\}$. We place a Gaussian prior on $\mathbf{w} = (w_0, w_1, w_2)$ with $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \alpha^{-1}\mathbf{I})$ where $\alpha = 2$. The observation noise variance is $\beta^{-1} = 0.1$. We fit the polynomial $y(x) = w_0 + w_1 x + w_2 x^2$.

(a) Posterior mean $\mathbf{m}_N$.

(b) Posterior covariance $\mathbf{S}_N$.

(c) Predictive mean and variance at $x_* = 1.5$.

Grading:

- *Posterior mean – 5 pts*
- *Posterior covariance – 5 pts*
- *Predictive distribution – 5 pts*

Total: 15 points

Model Evidence

Statement. We are comparing models using the Bayesian Marginal Likelihood (Model Evidence). We evaluate a model $\mathcal{M}$ with $M = 2$ parameters.

Assume a simplified 1D approximation for each parameter: before seeing data, the prior is completely flat with a width of $\Delta w_{\text{prior}} = 10$. After training, the posterior collapses into a tight uniform box of width $\Delta w_{\text{posterior}} = 0.1$. The likelihood of the training data evaluated at the MAP estimate is $p(\mathcal{D}|\mathbf{w}_{\text{MAP}}) = 5.0$.

(a) Compute the log-likelihood of the data at the MAP estimate.

(b) Compute the complexity penalty term for this model.

(c) Sum the above together to approximate log marginal likelihood $\ln p(\mathcal{D}|\mathcal{M})$.

Grading:

- *Log-likelihood evaluation – 2 pts*
- *Complexity penalty evaluation – 5 pts*
- *Total log evidence – 3 pts*

Total: 10 points

Evidence Approximation: Hyperparameter Updates

Statement. We are training a Bayesian Linear Regression model with $M = 2$ features and performing the iterative Evidence Approximation to find the optimal hyperparameters.

At the current iteration, we have: Prior precision $\alpha = 1.0$, Noise precision $\beta = 2.0$. The static eigenvalues of the data design matrix product $\mathbf{\Phi}^T\mathbf{\Phi}$ are exactly $\lambda_1 = 4.0$ and $\lambda_2 = 1.0$. The current MAP weight vector is $\mathbf{m}_N = (0.5, 1.0)^T$.

(a) Calculate the complexity parameter γ (the effective number of well-determined parameters).

(b) Perform the M-step to calculate the updated hyperparameter α_{new}.

Grading:

- *Complexity parameter γ – 5 pts*
- *Updated α_{new} – 5 pts*

Total: 10 points

Binary Classification via Least Squares Approximation

Statement. We have three data points in $\mathbb{R}^2$ with binary labels:

$$(x_1, t_1) = ((0,0), 0), \quad (x_2, t_2) = ((1,1), 1), \quad (x_3, t_3) = ((2,1), 1).$$

We attempt to fit them with a linear model

$$y(\mathbf{x}) = w_0 + w_1 x_1 + w_2 x_2$$

by minimizing the sum of squared errors $\sum_{n=1}^{3} [t_n - y(x_n)]^2$. Then we classify by thresholding at $y(\mathbf{x}) = 0.5$.

(a) Write down the design matrix $\mathbf{\Phi}$.

(b) Solve the normal equations for $\mathbf{w}$.

(c) Discuss the implications of thresholding at 0.5 and compare with logistic regression.

Grading:

- *Design matrix – 4 pts*
- *Solve normal equations – 6 pts*
- *Threshold vs logistic regression – 5 pts*

Total: 15 points

Problem 1: Polynomial Regression with Regularization (18 points)

Statement. We have four data points in 1D:

$$(x_1, t_1) = (0, 0), \quad (x_2, t_2) = (1,\, 0.8), \quad (x_3, t_3) = (2,\, 1.6), \quad (x_4, t_4) = (3,\, 3.7).$$

We fit the polynomial

$$y(x) \;=\; w_0 \;+\; w_1\, x \;+\; w_2\, x^2,$$

subject to a ridge (L2) penalty $\lambda = 1$:

$$\sum_{n=1}^{4} \big[y(x_n) - t_n\big]^2 \;+\; \lambda \sum_{j=0}^{2} w_j^2.$$

(a) Write the design matrix $\mathbf{\Phi}$.

(b) Formulate the ridge equation.

(c) Solve for (w_0, w_1, w_2) *numerically*, by explicitly writing and inverting the matrix.

Potentially useful inverses of matrices:

$$\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{pmatrix}^{-1} = \begin{pmatrix} 3 & -3 & 1 \\ -2.5 & 4 & -1.5 \\ 0.5 & -1 & 0.5 \end{pmatrix} \quad \begin{pmatrix} 5 & 6 & 14 \\ 6 & 15 & 36 \\ 14 & 36 & 99 \end{pmatrix}^{-1} = \begin{pmatrix} 0.3865 & -0.1840 & 0.0123 \\ -0.1840 & 0.6115 & -0.1963 \\ 0.0123 & -0.1963 & 0.0798 \end{pmatrix}$$

$$\begin{pmatrix} 4 & 6 & 14 \\ 6 & 14 & 36 \\ 14 & 36 & 98 \end{pmatrix}^{-1} = \begin{pmatrix} 0.95 & -1.05 & 0.25 \\ -1.05 & 2.45 & -0.75 \\ 0.25 & -0.75 & 0.25 \end{pmatrix}$$

Grading:

- *(a) Design matrix – 5 pts*
- *(b) Ridge equation form – 4 pts*
- *(c) Numeric solution with explicit matrix steps – 9 pts*

Total = 18 points

Problem 2: Multiclass Logistic Regression with Three Classes (17 points)

Statement. We have three classes $C_1,\ C_2,\ C_3$ and one 2D data point *from each class*:

$$\mathbf{x}_1 = (0,1)\ \in\ C_1, \quad \mathbf{x}_2 = (1,2)\ \in\ C_2, \quad \mathbf{x}_3 = (2,2)\ \in\ C_3.$$

Our *multiclass* logistic model has three activation functions:

$$a_1(\mathbf{x}) = w_{1,0} + w_{1,1}\,x_1 + w_{1,2}\,x_2, \quad a_2(\mathbf{x}) = w_{2,0} + w_{2,1}\,x_1 + w_{2,2}\,x_2, \quad a_3(\mathbf{x}) = w_{3,0} + w_{3,1}\,x_1 + w_{3,2}\,x_2.$$

(a) Suppose we have the parameter vectors:

$$\mathbf{w}_1 = (-1,\ 1,\ 0), \quad \mathbf{w}_2 = (-1,\ 0,\ 1), \quad \mathbf{w}_3 = (0,\ -1,\ 2).$$

Compute $a_1(\mathbf{x}_n),\ a_2(\mathbf{x}_n),\ a_3(\mathbf{x}_n)$ *and* the $\mathrm{softmax}\big(a_1, a_2, a_3\big)$ for each data point $n = 1, 2, 3$.

(b) Which class is predicted for each of the three points?

Grading:

- *(a) Softmax computations – 12 pts*
- *(b) Predicted class per point – 5 pts*

Total = 17 points

Problem 1: Sequential Gradient Descent in a Linear Regression Model

Statement. We have four one-dimensional data points:

$$(x_1, t_1) = (0,\ 0.1), \quad (x_2, t_2) = (1,\ 1.0), \quad (x_3, t_3) = (2,\ 1.9), \quad (x_4, t_4) = (3,\ 2.7).$$

We fit a linear model

$$y(x) \ = \ w_0 + w_1\, x,$$

where the sum-of-squares error function for all data is

$$E(\mathbf{w}) = \tfrac{1}{2} \sum_{n=1}^{4} \big[y(x_n) - t_n\big]^2.$$

The task is to perform *sequential* (online) gradient descent. We initialize $(w_0^{(0)}, w_1^{(0)}) = (0, 0)$, and we choose a fixed step size (learning rate) $\eta = 0.05$. Perform *one pass* over all 4 data points in the given order, showing each step.

(a) Write the partial derivatives $\frac{\partial E_n}{\partial w_0}$ and $\frac{\partial E_n}{\partial w_1}$ for a single data point (x_n, t_n), i.e. for $E_n = \frac{1}{2}\big(y(x_n) - t_n\big)^2$.

(b) Starting from $(0, 0)$, update (w_0, w_1) sequentially with step size $\eta = 0.05$ for the points in order: $(x_1, t_1), (x_2, t_2), (x_3, t_3), (x_4, t_4)$. Show the new (w_0, w_1) after each data point.

Grading:

- *(a) Formulation of derivatives – 6 pts*
- *(b) Sequential gradient descent – 12 pts*

Total = 18 points